**Scheme of Valuation**

**CIRCUIT SIMULATION AND PCB DESIGN (MG4SECECT200)**

**Part A – MCQ (35 × 1 = 35 Marks)**

- Correct answer: **1 mark**
- Wrong answer / No answer: **0 mark**

c) Ounces
b) Chemical process
d) All of these
b) Signal integrity
a) Microcontroller simulation
b) Noise reduction
b) Photoresist application
a) Reliability
c) Smooth curves
b) More control
c) Simple circuits
a) Copper
b) Design stage
c) Polyimide
b) PCB design
b) Compact design
b) Heat dissipation
b) Assembly error
b) Signal integrity
c) Schematic & PCB
c) SOIC
b) Proper spacing
a) Strong mechanical joint
b) Human error
b) Component leads
b) Wrong connections
a) HF signals
b) Manufacturing
b) Slow

d) All of these

c) Diode

b) Light

b) Connection mismatch

a) Noise

b) Fabrication

**Part B – Short Essay Questions (Any 3 × 5 = 15 Marks)**

**Q36. Advantages and Applications of Double-Sided PCBs (5 Marks)**

| Criteria | Marks |
|---|---|
| Definition/Introduction to double-sided PCB | 1 |
| Advantages (any four points) | 2 |
| Applications (any four applications) | 2 |

**Key Points**

- Components and tracks on both sides
- Higher circuit density
- Reduced board size
- Better routing flexibility
- Improved reliability
- Applications: consumer electronics, industrial control, power supplies, communication systems, automotive electronics.

**Q37. Through-Hole Plating Process and Role in Interconnection (5 Marks)**

| Criteria | Marks |
|---|---|
| Definition of through-hole plating | 1 |
| Process steps (drilling, cleaning, deposition, electroplating, inspection) | 2 |
| Role in interlayer connection | 1 |
| Importance/reliability/mechanical strength | 1 |

**Q38. Workflow for Schematic and PCB Layout Using EasyEDA (5 Marks)**

| Criteria | Marks |
|---|---|
| Project creation and schematic drawing | 1 |
| Component selection and footprint assignment | 1 |
| ERC and conversion to PCB layout | 1 |
| Placement and routing | 1 |
| DRC, Gerber generation and fabrication output | 1 |

**Q39. EMI and its Effects on PCB Performance (5 Marks)**

| Criteria | Marks |
|---|---|
| Definition of EMI | 1 |
| Effects on PCB performance (any four points) | 2 |
| EMI reduction methods (any four methods) | 2 |

**Effects**

- Signal distortion
- Data errors
- Reduced reliability
- Analog noise
- Device malfunction

**Reduction Techniques**

- Ground planes
- Proper spacing
- Shielding
- Filtering
- Controlled impedance routing

**Q40. SMT Assembly Process with Examples (5 Marks)**

| Criteria | Marks |
|---|---|
| Definition of SMT | 1 |
| SMT process steps | 2 |
| Examples of SMT components | 1 |
| Advantages/applications | 1 |

**Process Steps**

1. Solder paste application
2. Pick-and-place assembly
3. Reflow soldering
4. Inspection and testing

**Examples**

- SOIC ICs
- SMD resistors
- SMD capacitors
- SMD LEDs

**Applications**

- Smartphones
- Computers
- Consumer electronics
- Communication devices